Patterns

(Class Slides)

Note: We are sharing these slides as lot of mathematical logic building was done on paper.

Hope these will help you in your learning process

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```
1 1 1 1
2 2 2 2
3 3 3 3
4 4 4 4
```

```
*
**
***
****
```

```
****
***
**
*
```

```
1
12
123
1234
```

```
A
BC
DEF
GHIJ
```

```
*****
*   *
*   *
*****
```

```
    *
   **
  ***
 ****
```

```
1
2  3
4  5  6
7  8  9  10
11 12 13 14 15
```

```
   *
  ***
 *****
*******
*******
 *****
  ***
   *
```

```
*       *
**     **
***   ***
*********
*********
***   ***
**     **
*       *
```

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Nested Loops

Loop inside a loop

$n = 4$

n

	C1	C2	C3	C4
Row1 →	1	1	1	1
R2 →	2	2	2	2
R3 →	3	3	3	3
R4 →	4	4	4	4

```
for(int i=1; i<=4; i++){
    for(int j=1; j<=4; j++){
        cout << i << " ";
    }
}
```

1. Outer loop : no. of Rows
2. Inner loop : no. of columns / each row
3. Work in inner loop (each row)
   i print

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Print Star pattern

$n = 4$

R1 * 1st
R2 ** 2 st
R3 *** 3 st
R4 **** 4 st

① outer loop → Rows (i) → Row no.
n times (1 to n)

② inner loop (each rows/columns)
i times (1 to i)

③ work ?
cout<<"*"

```
for (int i=1; i<=n; i++) {
    for (int j=1; j<=i; j++) {
        cout << "*"
    }
    cout << endl;
}
```

n = 4

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Print Inverted Star pattern

R1 * * * *
R2 * * *
R3 * *
R4 *

$n = 4$

$4 = n - i + 1$
$3 = n - i + 1$
$2 = n - i + 1$
$1 = n - i + 1$

1. outer loop (rows)
   (1 to n)
2. inner loop (each row)
   (1 to n-i+1)
3. work?
   cout << "*"

n = 4

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Print Half Pyramid pattern

$n = 4$

R1	1	1
R2	12	1 to 2
R3	123	1 to 3
R4	1234	1 to 4

1 to i

```
for(int i=1; i<=n; i++) {
    for(int j=1; j<=i; j++) {
        cout<<j;
    }
    endline
}
```

① outer loop (rows)
(1 to n)

② inner loop (each row)
(1 to i) ⇒ j

③ work?

```
cout<<j;
```

n = 4

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APNA COLLEGE

Print Character Pyramid pattern

A
BC
DEF
GHIJ

n = 4

n = 4

char ch = 'A'

① outer loop (row)
(1 to n)

② inner loop (each row)
(1 to i)

③ work ?

```
cout << ch;
ch++; //
```

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Print Hollow Rectangle pattern

R1	* * * * *	1st + 3st + 1st
R2	* _ _ _ *	1st + 3sp + 1st
R3	* _ _ _ *	1st + 3sp + 1st
R4	* * * * *	1st + 3st + 1st

first or last → 5 stars
(1) (n)

① Outer loop (rows)
(1 to n)

② Inner loop (each row)

```
cout << "*"; //First
for ( 1 to n-1) {
    1st or last → "*"
    else → " "
}
cout << "*"; //Last
```

③ work

n = 4

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$n = 4$

Inverted & Rotated Half-Pyramid

_ _ _ *	R1	3 sp	+ 1st
_ _ * *	R2	2 sp	+ 2st
_ * * *	R3	1 sp	+ 3st
* * * *	R4	0 sp	+ 4st

	n − i spaces
$i=1$	$4-1 = 3$
$i=2$	$4-2 = 2$
$i=3$	$4-3 = 1$
$i=4$	$4-4 = 0$

n = 4

① outer loop (rows)
(1 to n)

② inner loop (each row)

a) Spaces (1 to n−i)
cout << " " ← work

b) Stars (1 to i)
cout << "*" ← work

cout << endl

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Print Floyd's Triangle

num = 1

R1	1	1 el
R2	2 3	2 el
R3	4 5 6	3 el
R4	7 8 9 10	4 el
R5	11 12 13 14 15	5 el

ith

i times
(1 to i)

① outer loop (rows)
(1 to n)

② inner loop (each row elements)
(1 to i)

③ work ?

```
cout << num;
num++;
```

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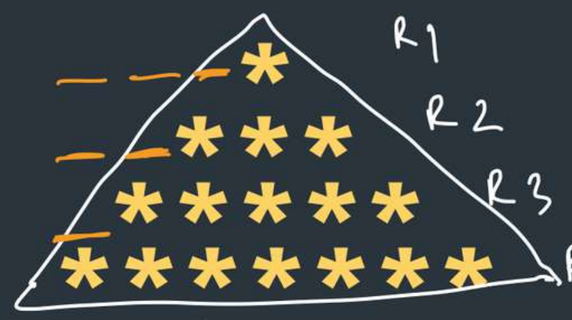
Diamond Pattern

$n = 4$



3sp + 1st
2sp + 3st
1sp + 5st
0sp + 7st

1st pyramid

① outer loop (rows) (1 to n)

② inner loop (each row elements)

a) (1 to n-i) cout << " "

b) (1 to 2*i-1) cout << "*"

i=1 → 1 $\quad 2*i-1 = 1$

i=2 → 3 $\quad 2*i-1 = 3$

i=3 → 5 $\quad 2*i-1 = 5$

i=4 → 7 $\quad 2*i-1 = 7$

n = 4

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Diamond Pattern

$n = 4$

```
*******   R4
-*****    R3
--***     R2
---*      R1
```

0 sp	+	7 st
1 sp	+	5 st
2 sp	+	3 st
3 sp	+	1 st
		$2*i-1$

2nd pyramid

1. outer loop (rows) (n to 1)
2. inner loops (each row elements)

a) sp (1 to $n-i$)

b) st (1 to $2*i-1$)

n = 4

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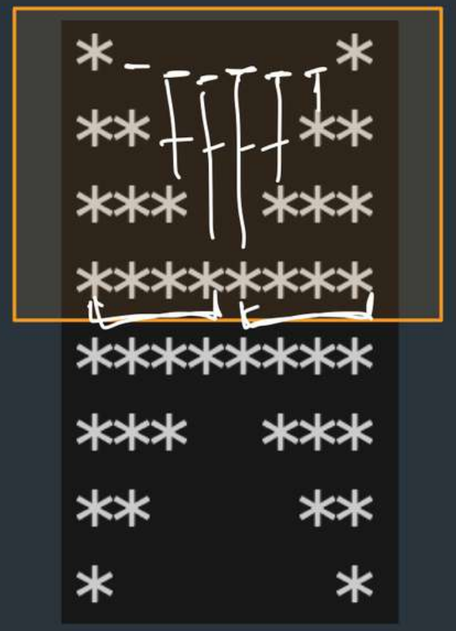
Print Butterfly Pattern

$n = 4$

```
*       *
**     **
***   ***
********
********
***   ***
**     **
*       *
```



R1 : 1st + 6sp + 1st

R2 : 2st + 4sp + 2st

R3 : 3st + 2sp + 3st

R4 : 4st + 0sp + 4st

$2 * (n - i)$

Pattern1

① outer loop (rows) (1 to n)

② inner loop

a) stars (1 to i)

b) spaces (1 to 2*(n-i))

c) stars (1 to i)

n = 4

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APNA COLLEGE

Print Butterfly Pattern

n=4

```
*       *
**     **
***   ***
********
********
***   ***
**     **
*       *
```

```
*       *
**     **
***   ***
********
********
***   ***
**     **
*       *
```

R4 4 st 0 sp ✓ 4st
R3 3 st 2 sp ✓ 3st
R2 2 st 4 sp ✓ 2st
R1 1 st 6 sp ✓ 1st

$2*(n-i) = 2*(4-1) = 6$

$i = 1$

n = 4

Pattern

① outer loop (rows) (n to 1)

② inner loop

a) stars (1 to i)

b) spaces (1 to 2*(n-i))

c) stars (1 to i)

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